Prompted Segmentation for Drywall QA

Goal

Fine-tune a single text-conditioned segmentation model that, given an image and a natural-language prompt, produces a binary mask for:

- **"segment crack"** (cracks dataset)
- **"segment taping area"** (drywall-join dataset)

Approach

Model: CLIPSeg (`CIDAS/clipseg-rd64-refined`), a CLIP-based model with a lightweight decoder for text-prompted segmentation.

Why CLIPSeg: The assignment requires a text-conditioned interface (`image + prompt -> mask`). CLIPSeg is a natural (and relatively lightweight) choice, where one model handles both tasks via the prompt, with no task-specific heads or separate models needed.

Fine-tuning strategy:

- Freeze the CLIP text encoder entirely
- Freeze most of the CLIP vision encoder
- Unfreeze the last 2 vision encoder layers for domain adaptation
- Train the segmentation decoder
- This gives 15.7M trainable parameters out of 150.7M total (~10.4%)

Training details:

- **Loss:** Weighted BCE + Dice loss, with per-task positive-pixel weighting (cracks are sparse) and per-task sample weighting (drywall labels are weaker since they are just bounding boxes)
- **Augmentation:** Random horizontal flip (50%), vertical flip (20%), rotation +/-15 deg (30%), color jitter (brightness, contrast, saturation)
- **Prompt variants during training:** `"segment crack"`, `"segment wall crack"` for cracks; `"segment taping area"`, `"segment joint/tape"`, `"segment drywall seam"` for drywall. These are taken from the task pdf.
- **LR schedule:** Cosine decay with 5% linear warmup
- **Optimizer:** AdamW (lr=1e-4, weight_decay=1e-4)
- **AMP:** Enabled (fp16 on CUDA)
- **Gradient clipping:** max_norm=1.0
- **Early stopping:** patience=5 (did not trigger; ran full 20 epochs)
- **Post-training threshold search:** Per-task optimal threshold selected on validation set (cracks=0.60, drywall=0.45)
- **Seed:** 7

Datasets

	Cracks	Drywall-Join-Detect
Source	Roboflow	Roboflow
Images	5,000	1,022
Annotations	7,930	1,424
Image size	640x640	640x640
Mask type	Polygon segmentation (99.5%)	**Bounding box only**
Canonical prompt	`"segment crack"`	`"segment taping area"`

Note: The drywall dataset provides only bounding boxes, not segmentation masks. The model trains on

bbox-derived pseudo-masks for this task, not on more refined segmented masks. One option to get segmented masks is to run SAM on the bounding boxes, but this was not done primarily due to time constraints, but also because SAM would introduce at least some errors, which would lead to corruption of the ground truth.

Data Splits

Splits are created deterministically (seed=7) with each dataset split separately to prevent the smaller drywall set from being drowned out.

Split	Cracks	Drywall	Total
Train (70%)	3,500	715	4,215
Val (15%)	750	153	903
Test (15%)	750	154	904

Results

Baseline vs Fine-Tuned (Test Set)

Task	Model	IoU	Dice	IoU Gain
Cracks	Zero-shot CLIPSeg	0.2288	0.3383	—
Cracks	**Fine-tuned CLIPSeg**	**0.4898**	**0.6359**	**+0.2610**
Drywall	Zero-shot CLIPSeg	0.0271	0.0384	—
Drywall	**Fine-tuned CLIPSeg**	**0.5445**	**0.6924**	**+0.5174**
Macro	Zero-shot CLIPSeg	0.1279	0.1883	—
Macro	**Fine-tuned CLIPSeg**	**0.5171**	**0.6641**	**+0.3892**

Consistency (Test Set)

Task	IoU Mean	IoU Std	IoU Min	Dice Mean	Dice Std	Dice Min
Cracks	0.4898	0.1821	0.0100	0.6359	0.1785	0.0198
Drywall	0.5445	0.1428	0.0000	0.6924	0.1391	0.0000

- Drywall predictions are more consistent (lower std) despite training on weaker bbox labels.
- Cracks have higher variance, driven by thin/faint cracks that the model struggles with.
- Even though drywall had less examples and lesser quality masks, the model seems to perform better on it because the ground truth is easier to learn (rectangular bbox/straight lines instead of curved cracks)
- A small number of images have near-zero scores

Optimal Thresholds (from validation search)

Task	Threshold
Cracks	0.60
Drywall	0.45

These were found by sweeping 0.10–0.80 on the validation set and selecting the threshold that maximizes per-task IoU. Test evaluation uses these per-task thresholds.

Training Curves

Loss

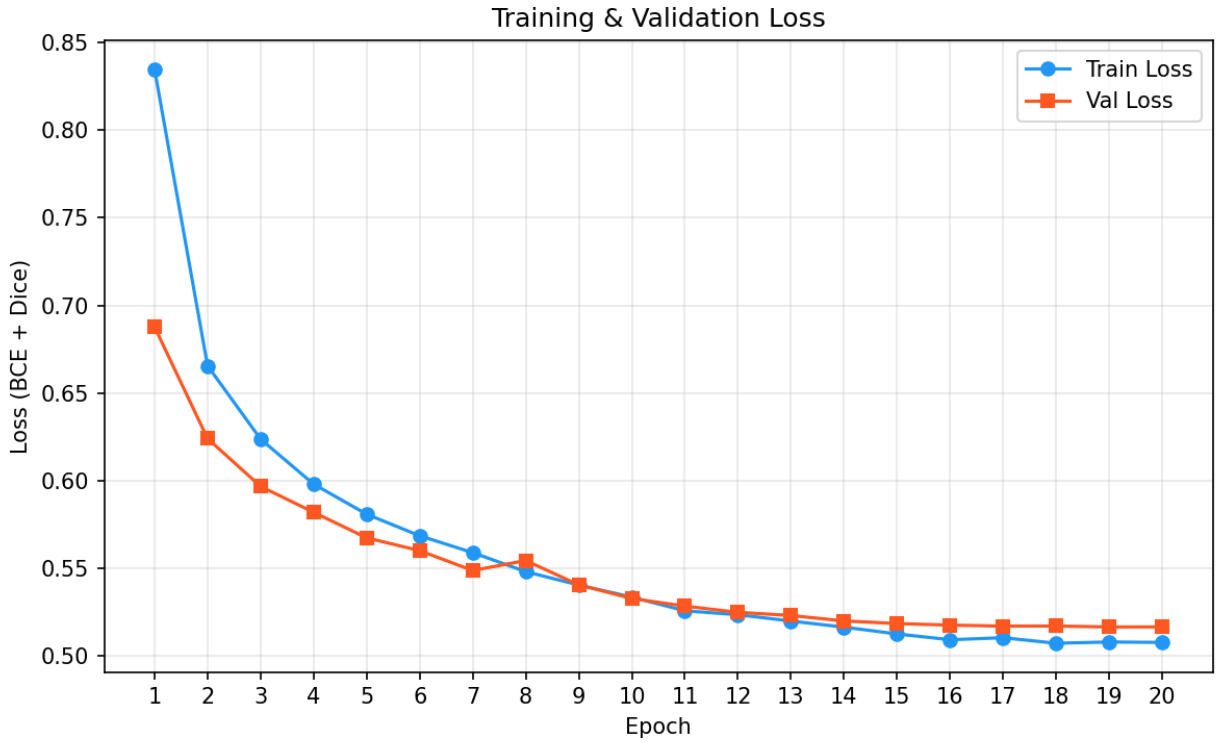


Figure 1: Training & Validation Loss

Macro IoU & Dice

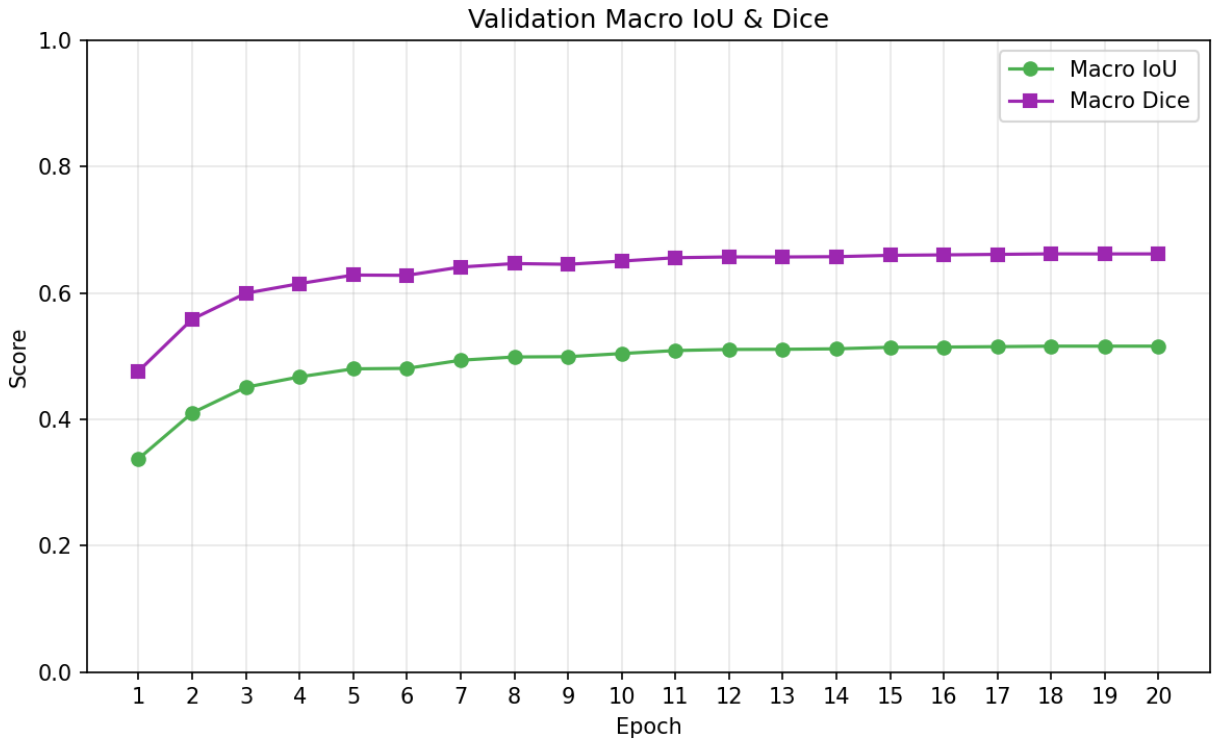


Figure 2: Macro Metrics

Per-Task Metrics

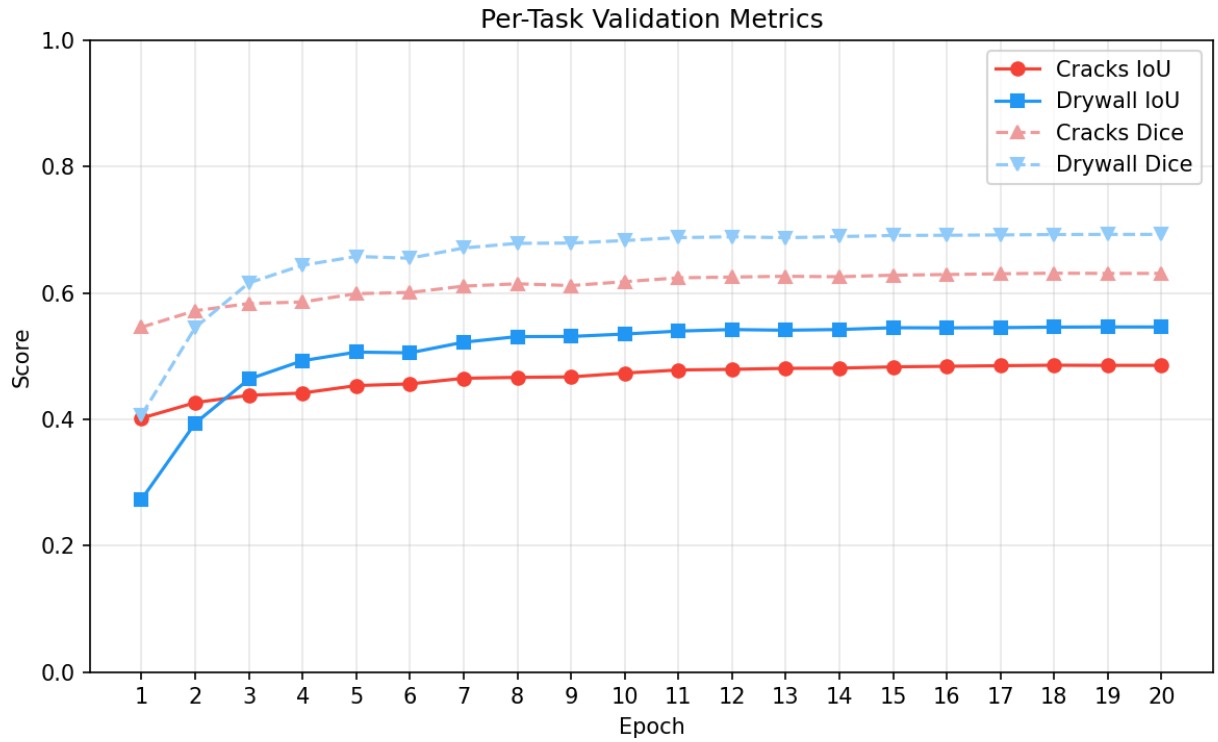
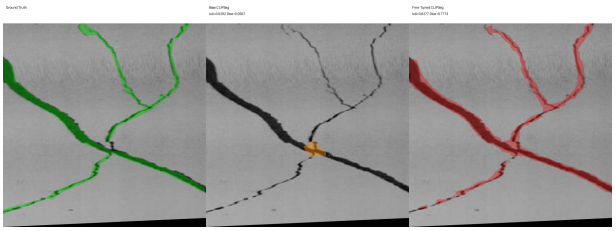


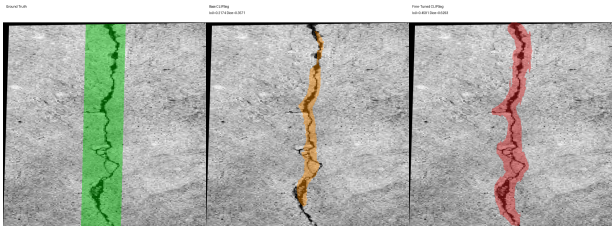
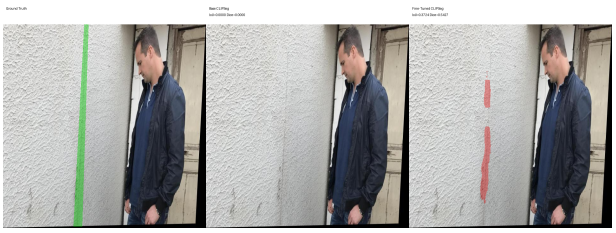
Figure 3: Per-Task Metrics

Training converges around epoch 15–16. The model trained for all 20 epochs (early stopping patience of 5 was not triggered). Best checkpoint selected by validation macro IoU.

Visual Examples

Crack Examples

Example	Description
	The finetuned model performs much better than the base model and is able to predict the complex geometry of this crack

Example	Description
	Here, the finetuned model seems to perform much better than even the ground truth segmentation! The GT mask is just a bbox but the finetuned model produces a much finer mask
	Here, the model seems to fail to perform well. It can be due to the complexity of the scene due to which the crack is not really the most noticeable

Drywall Examples

Example	Description
	We see that the finetuned model predicts a decent mask, where the base model didn't predict anything at all
	Here also the finetuned model performs nicely. We can see that the mask shape is a bit irregular (in the previous example also) because the ground truth is a (relatively) big bounding box so it is not very precise in capturing just the drywall taping area

Example	Description
	Here the finetuned model performs bad. It could be due to the complexity of the scene where we have a lot of random black boxes and also because the target area seems quite faint

Failure Analysis

- **Thin/faint ground truths:** CLIPSeg's decoder operates at reduced resolution (~352x352) and upsamples to 640x640. Very thin cracks or drywalls can be lost during this process, leading to near-zero IoU on some images.
- **Drywall zero-score images:** A few drywall test images produce completely empty predictions. These may be images where the joint is subtle or where the bbox pseudo-mask training signal was too weak. In some drywall examples, there are random black squares also. This can make the prediction more difficult.
- **Drywall label noise:** All drywall ground truth is derived from bounding boxes, which overestimate the true mask area. The reported IoU/Dice for drywall is measured against these box pseudo-masks, not against true segmentation masks.

Runtime & Footprint

Metric	Value
Total parameters	150.7M
Trainable parameters	15.7M (10.4%)
Model size (safetensors)	576 MB
Training time	2,341s (~39 min)
Training hardware	NVIDIA RTX 3050 Ti (4GB VRAM)
Avg inference time	54.9 ms/image
Batch size (training)	4
Epochs trained	20

Prediction Masks

Prediction masks are saved as single-channel PNGs with pixel values {0, 255}, matching the source image spatial size (640x640). Filenames follow the format:

```
<image_id>__segment_crack.png
<image_id>__segment_taping_area.png
```

See `outputs/predictions/` for the full test set predictions.

How to Reproduce

Setup

```
python3 -m venv venv
source venv/bin/activate
```

```
pip install torch transformers pillow numpy matplotlib
```

Train

```
python3 scripts/finetune_clipseg.py \
  --output-dir runs/clipseg_v2 \
  --epochs 20 \
  --batch-size 4 \
  --learning-rate 1e-4 \
  --unfreeze-vision-layers 2 \
  --early-stopping-patience 5 \
  --seed 7
```

Inference (generate prediction masks)

```
python3 scripts/inference.py \
  --run-dir runs/clipseg_v2 \
  --output-dir outputs/predictions \
  --split test
```

Compare baseline vs fine-tuned

```
python3 scripts/compare_clipseg_models.py \
  --run-dir runs/clipseg_v2 \
  --split test \
  --save-dir outputs/clipseg_v2_compare
```

Plot training curves

```
python3 scripts/plot_training.py \
  --run-dir runs/clipseg_v2 \
  --output-dir figures
```

Repository Layout

```
.
   README.md
   Prompted_Segmentation_for_Drywall_QA.pdf
   scripts/
       finetune_clipseg.py          # Main training script
       inference.py                  # Prediction mask generation
       compare_clipseg_models.py     # Base vs fine-tuned comparison
       plot_training.py              # Training curve plots
       zero_shot_clipseg_baseline.py # Zero-shot baseline
       grounded_sam_baseline.py      # Grounded SAM baseline
   cracks.coco/                       # Crack segmentation dataset
   Drywall-Join-Detect.coco/          # Drywall joint detection dataset
   runs/                              # Training outputs
       clipseg_v2/
           run_config.json
           history.json
           best_thresholds.json
           test_metrics.json
           best_model/
   outputs/                           # Predictions and comparisons
```

```
    predictions/
    clipseg_v2_compare/
figures/                          # Training curve plots
```